



Recovering Socioeconomic Information from AlphaEarth Embeddings

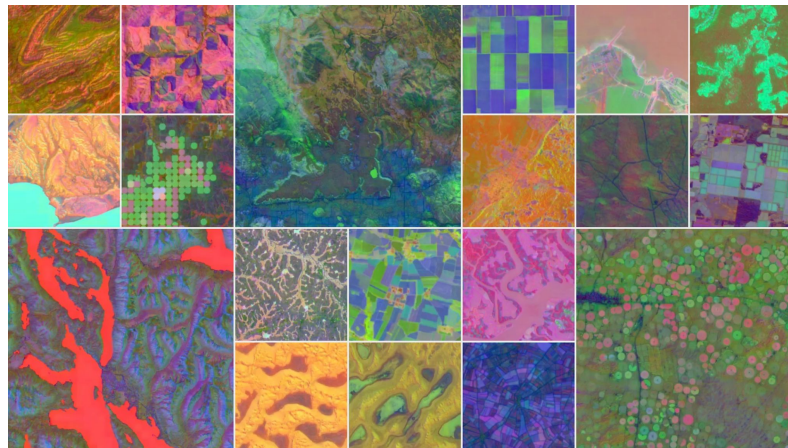
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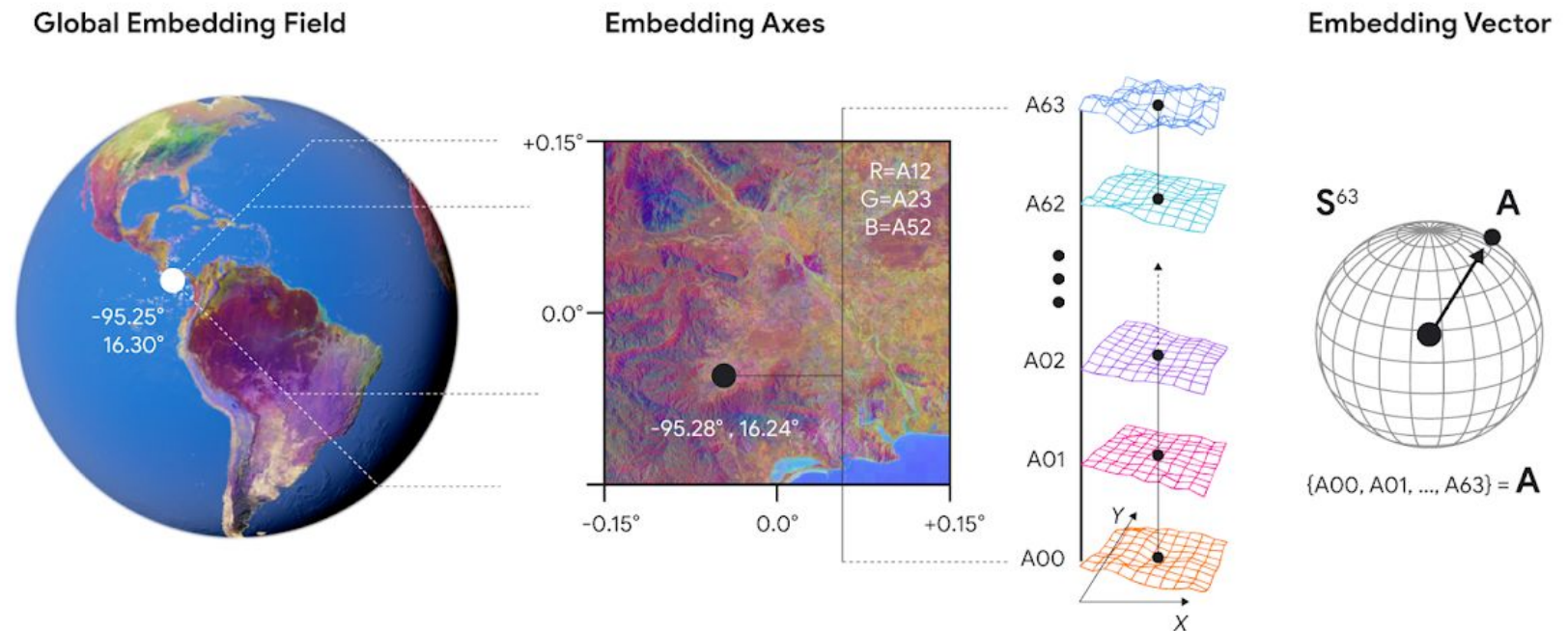
Motivation

- Global pretrained embeddings capture rich semantics
- Google released a global embedding space (**AlphaEarth**) that AI can process easily
- How can we use this in existing ML models?
- How does this affect model performance and generalization?



Background

- 64-dimensional embedding vector normalized to 1
- Inputs include optical, radar, LiDAR, environmental, and text data

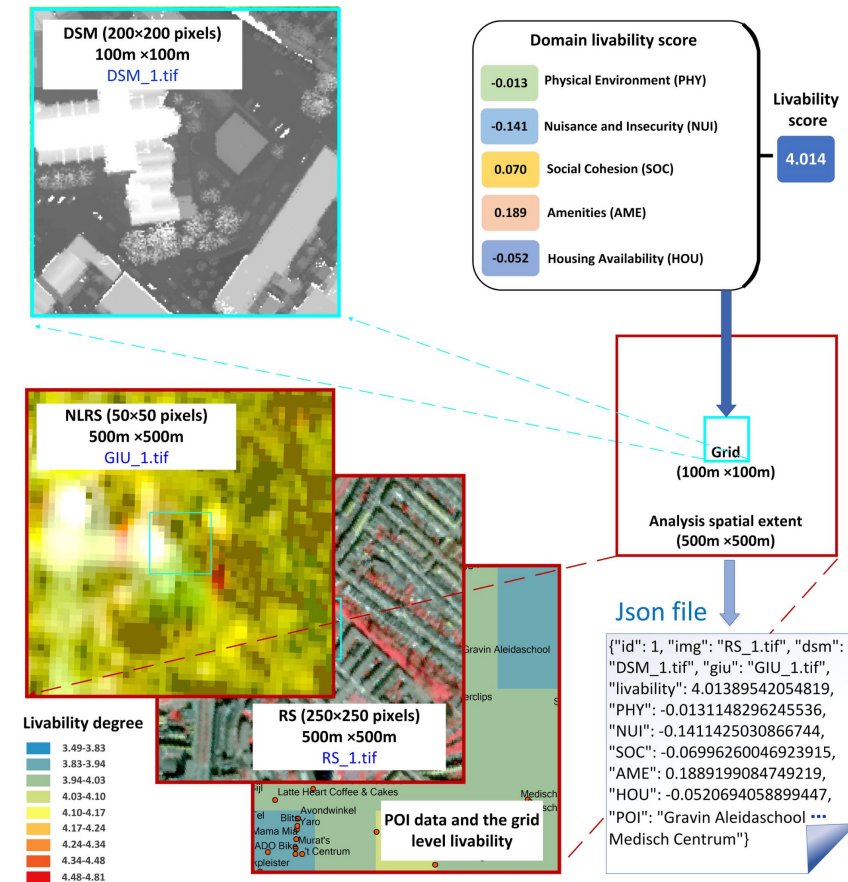


Problem

- Livable cities promote economic growth, physical/mental health, and sustainability
- Mainstream methods of evaluating livability include assigning weights to a variety of indicators
- Zhou et al. (2026) utilize a transformer-based model based on geospatial data to evaluate urban livability
 - Select certain inputs known to derive information on urban livability, requiring data preprocessing

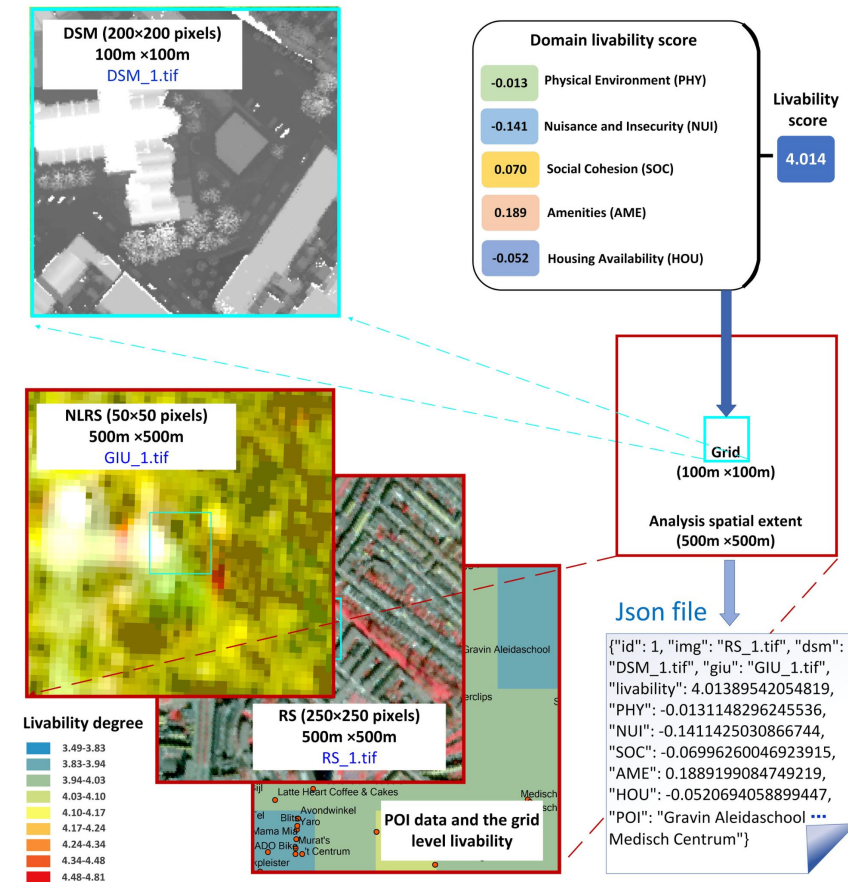
Datasets

- Inputs:
 - Remote Sensing (RS)
 - Digital Surface Model (DSM)
 - Night Light Remote Sensing (NLRs)
 - Point of Interest (POI) data
 - AlphaEarth Embeddings



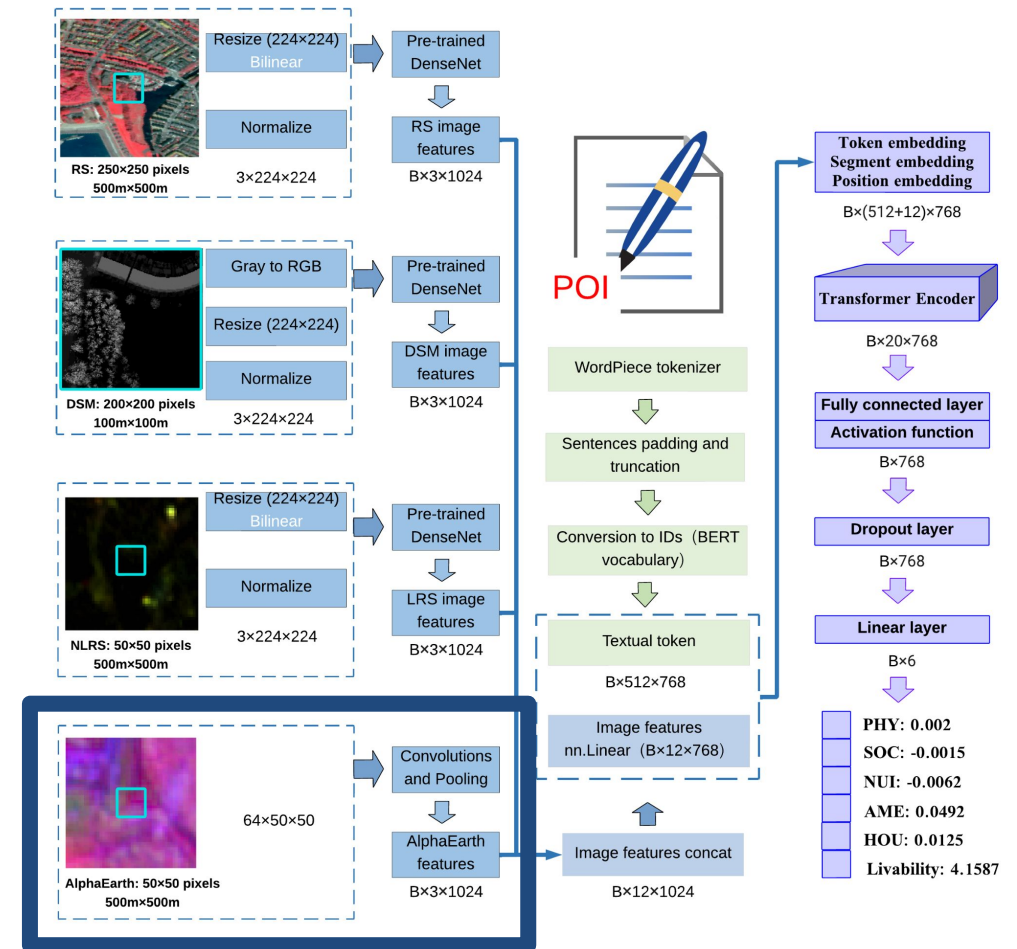
Datasets

- Outputs:
 - Overall Livability (LIV)
 - Physical Environment (PHY)
 - Nuisance and Safety (NUI)
 - Social Cohesion (SOC)
 - Housing Availability (HOU)
 - Amenities (AME)



Model

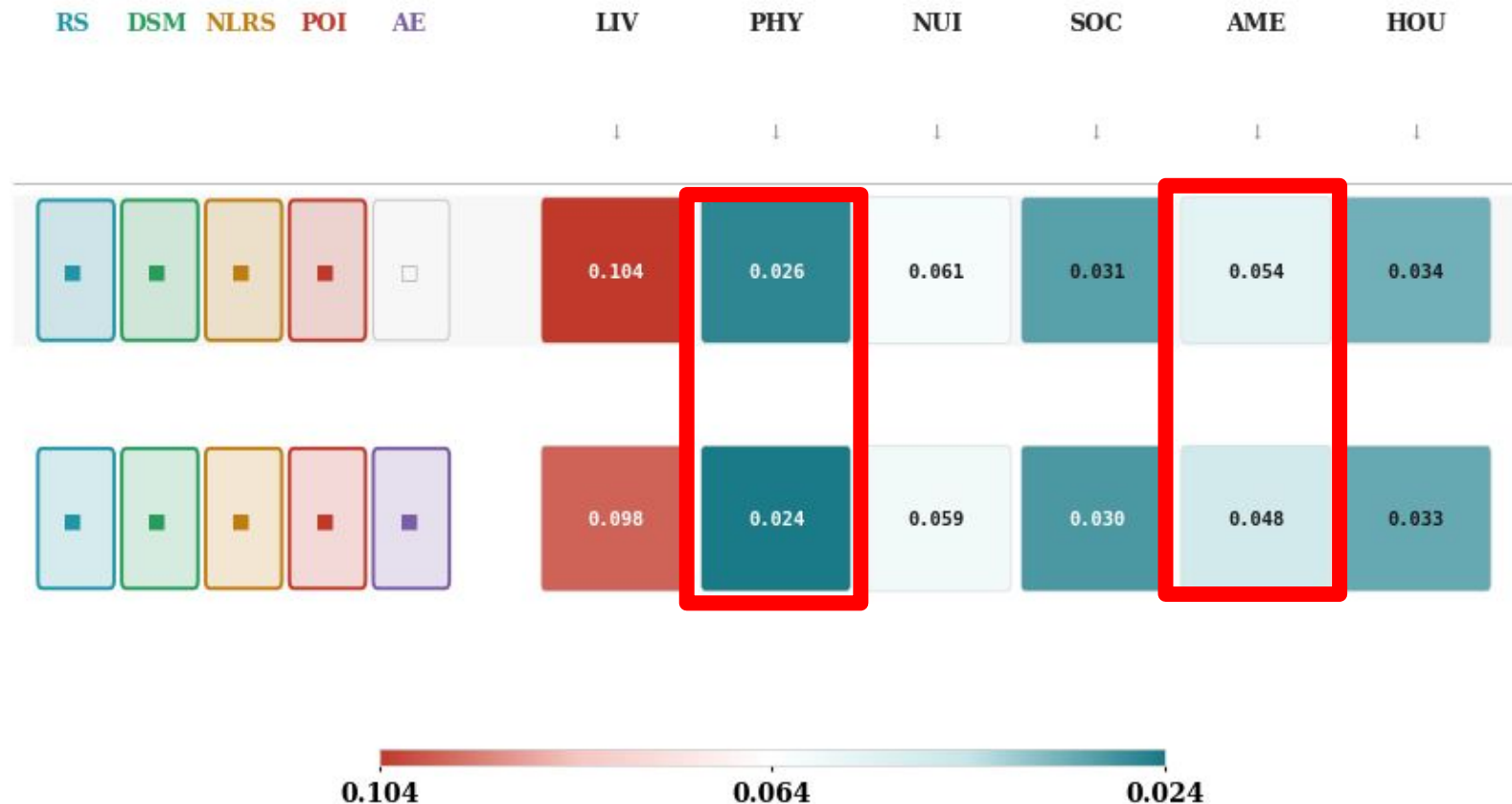
- Alter original transformer arch
- We process AlphaEarth through:
 - convolutions and pooling
 - concatenation with embeddings





RMSE

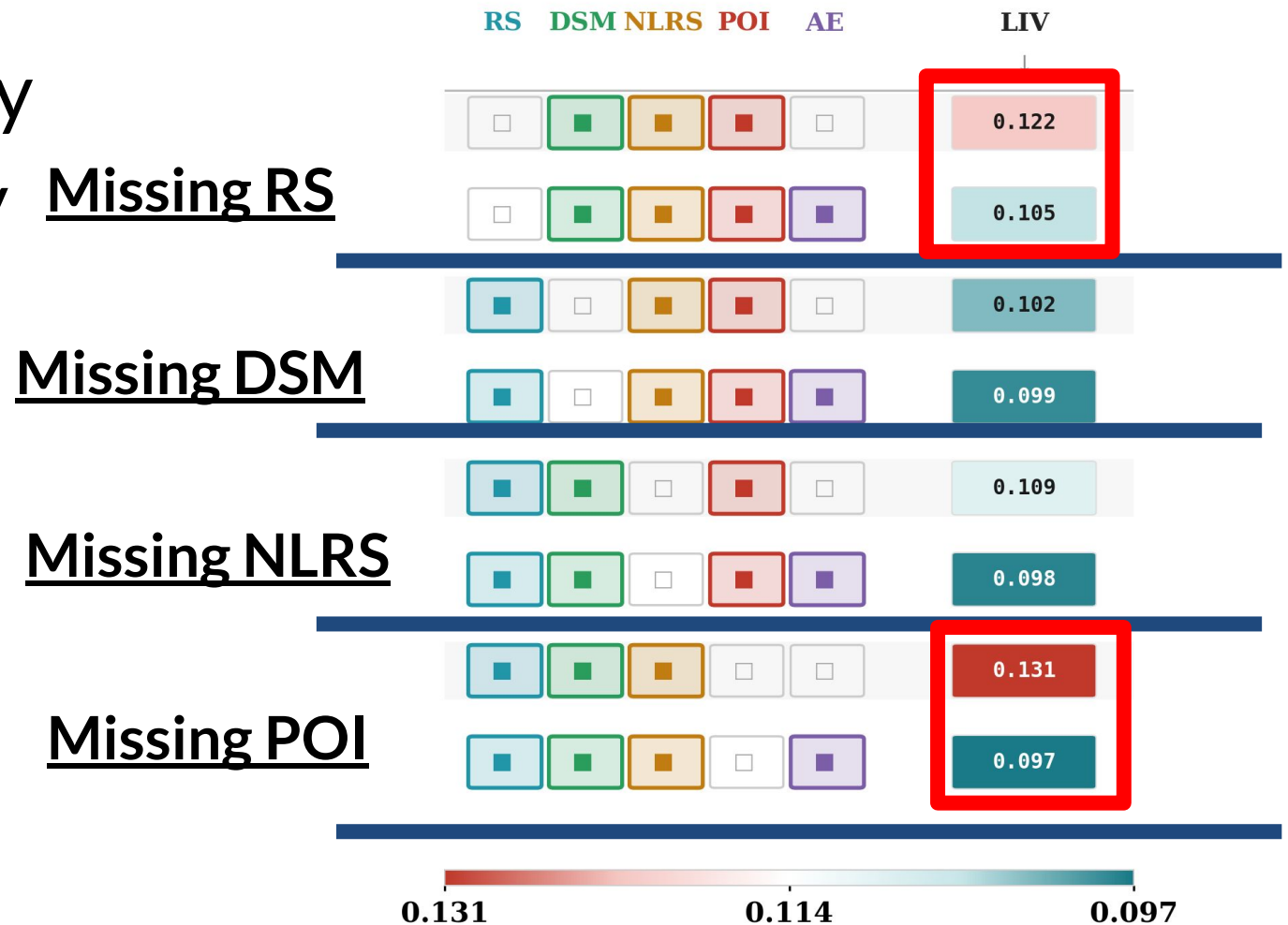
RMSE of **AME** and **PHY** both decrease by 11% and 8%



RMSE by Modality Combinations

Adding AlphaEarth greatly improved overall livability performance when there was no

- Remote Sensing
- Point of Interest

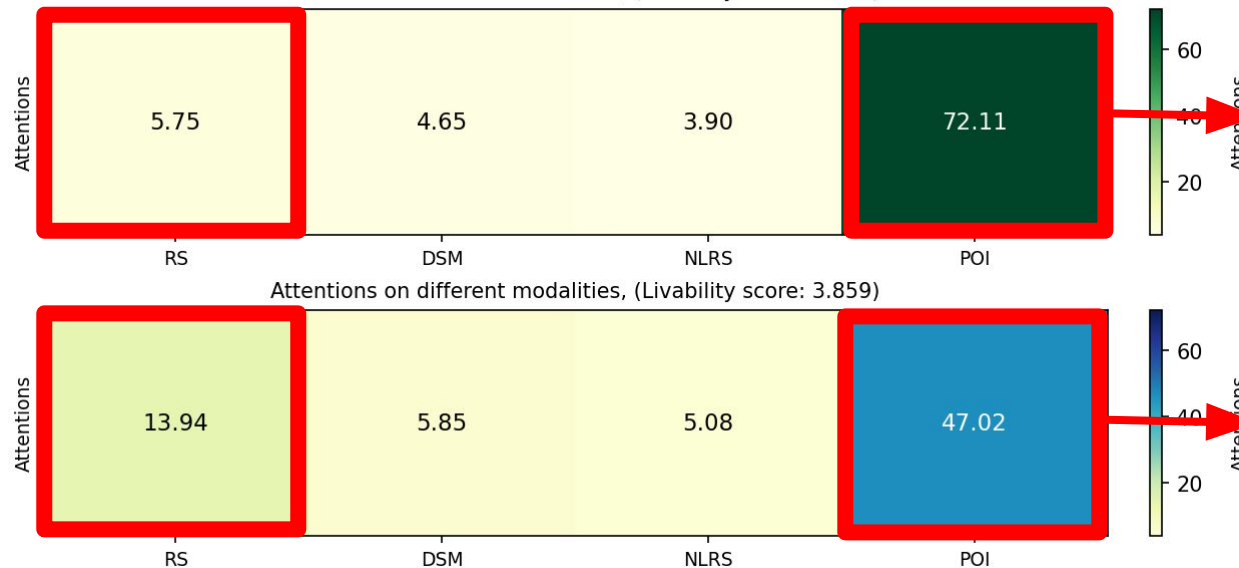


Attention Scores

- Adding AlphaEarth boosts the attention of DSM and NLRS while decreasing the attention of RS and POI
- Strong focus on POI data is likely due to heterogeneity

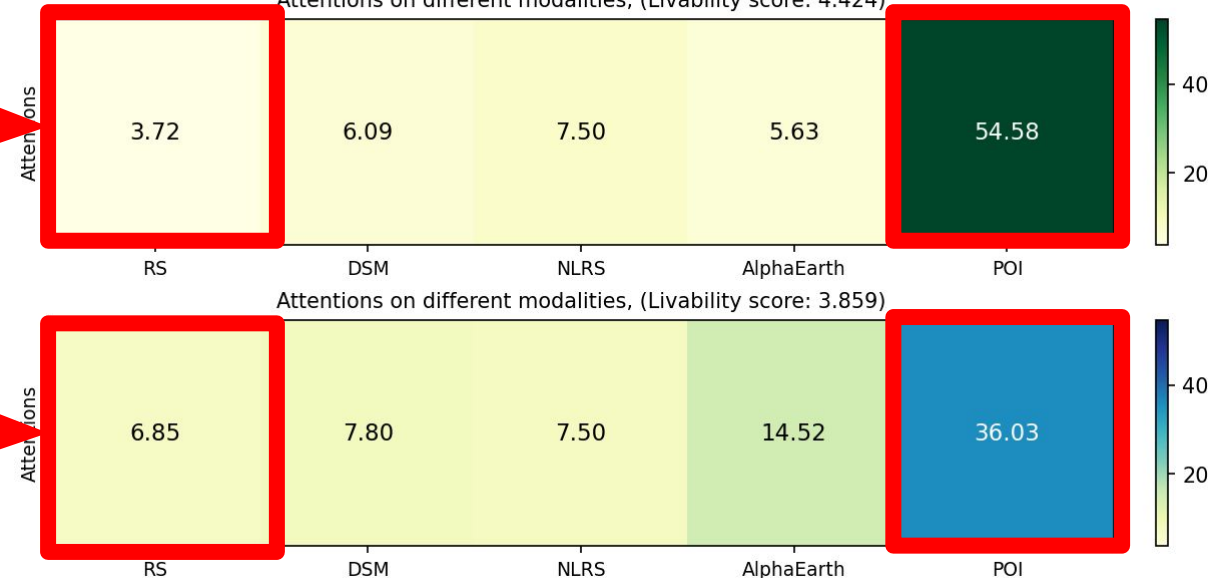
Baseline

Attentions on different modalities, (Livability score: 4.424)

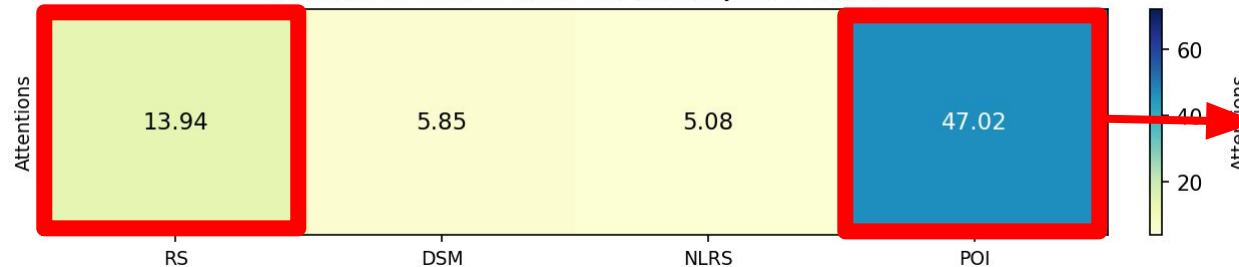


AlphaEarth

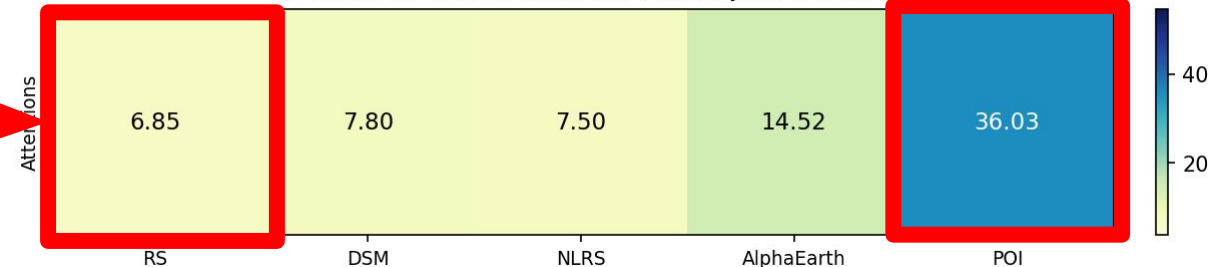
Attentions on different modalities, (Livability score: 4.424)



Attentions on different modalities, (Livability score: 3.859)



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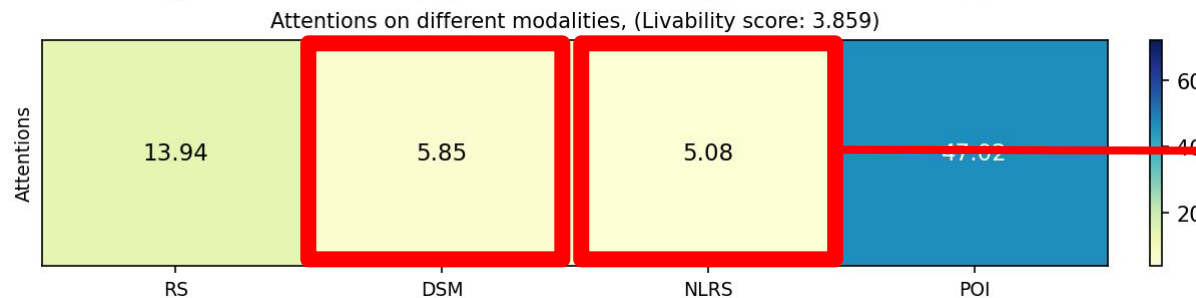
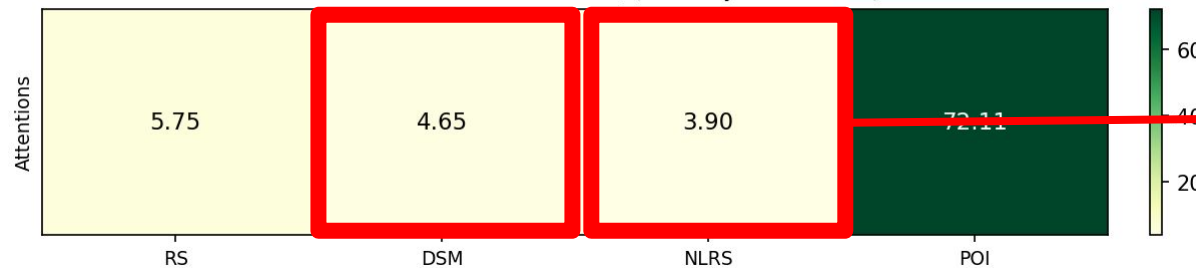


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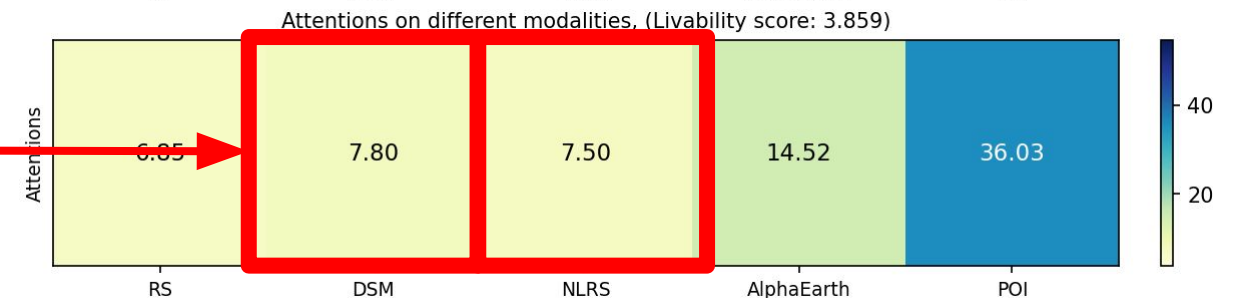
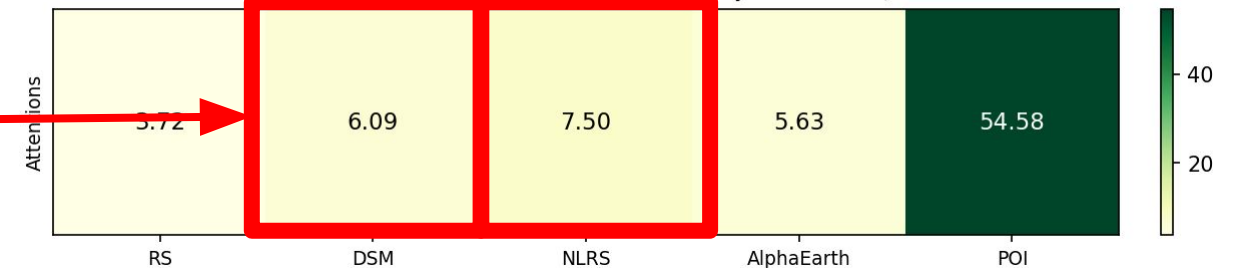
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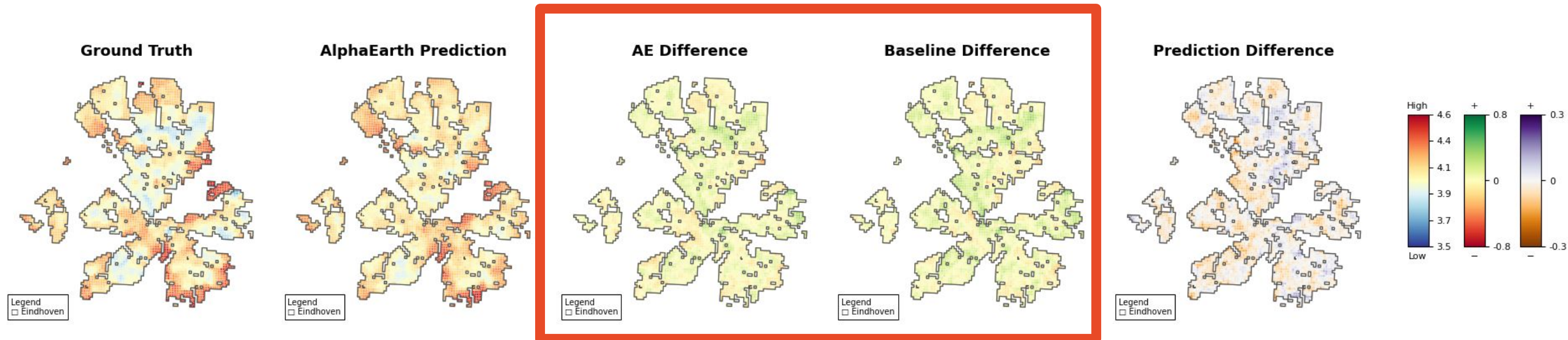
AlphaEarth

Attentions on different modalities, (Livability score: 4.424)



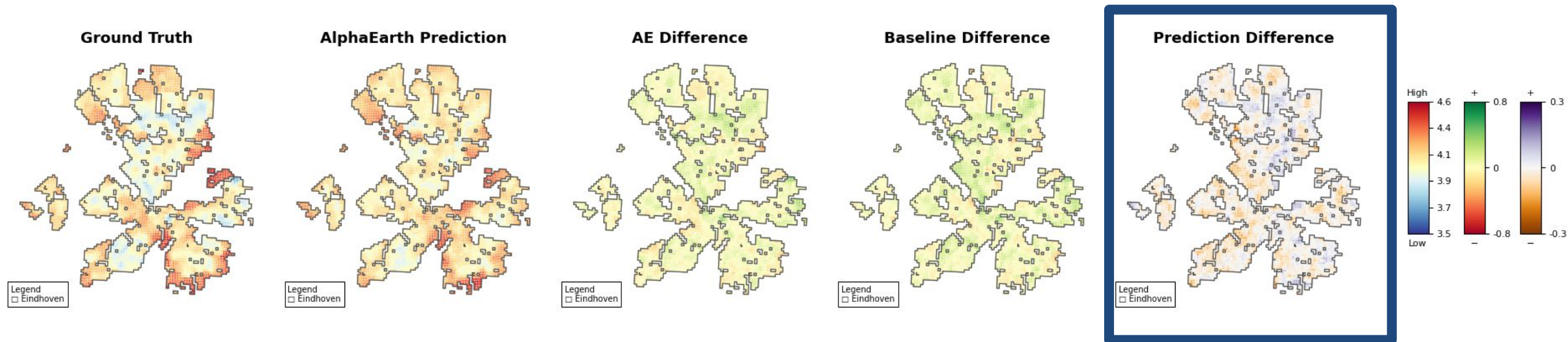
Livability Plots

- Both overestimate livability in Eindhoven
 - Green = Overestimation of livability
 - Orange = Underestimation of livability



Livability Plots

- AlphaEarth tends to produce slightly lower livability scores compared to the baseline model
 - Orange = AlphaEarth scores are lower than baseline
 - Purple = AlphaEarth scores are higher than baseline



Conclusions and Future Directions

- AlphaEarth consistently improves livability prediction performance by unifying modalities
- AlphaEarth can effectively replace certain input modalities during training, reducing the need for hard-to-obtain data sources
- Are there certain channels of AlphaEarth that correspond to certain information? How can we find them out?

Acknowledgements

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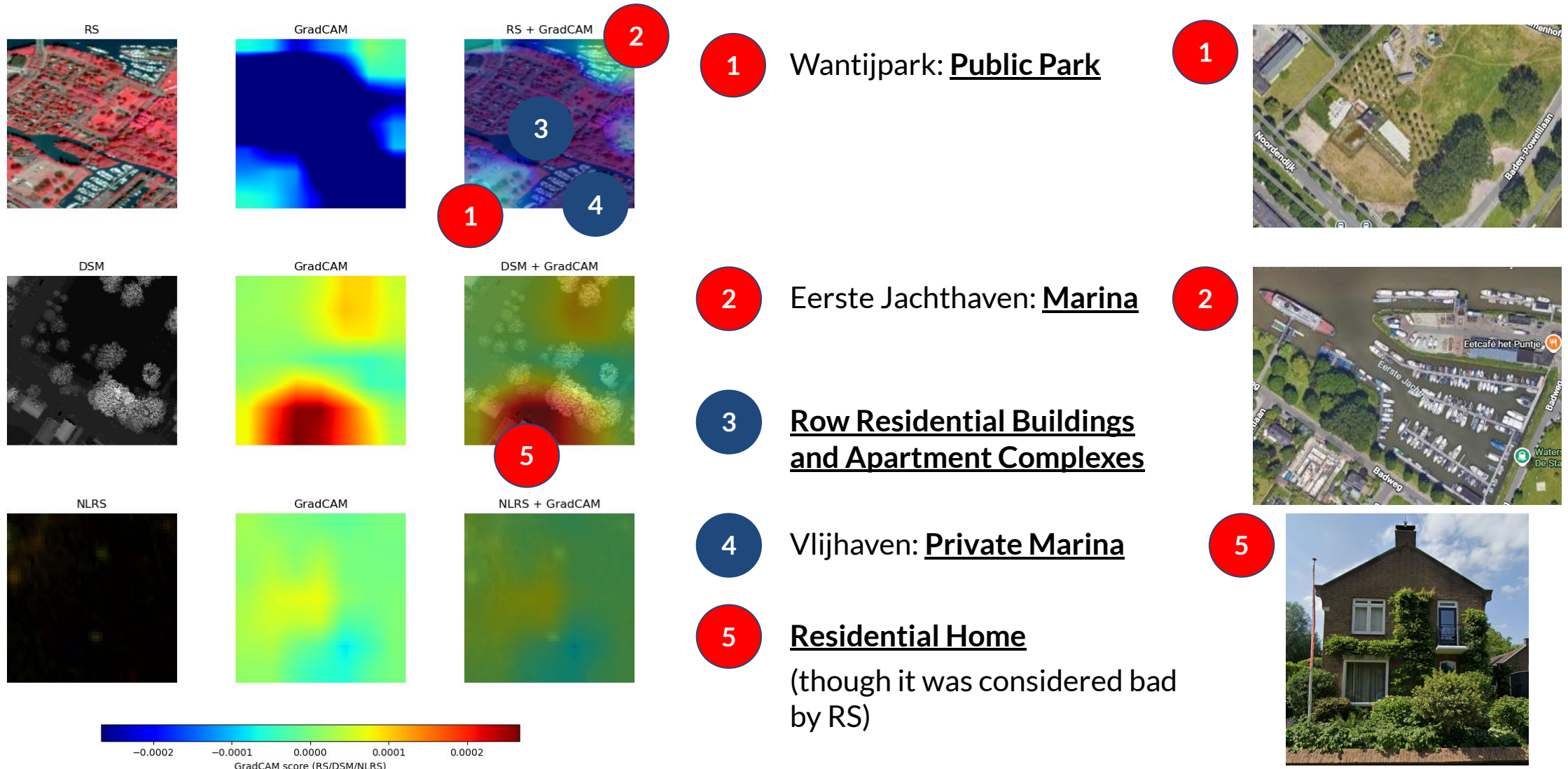


Thank you for your time!

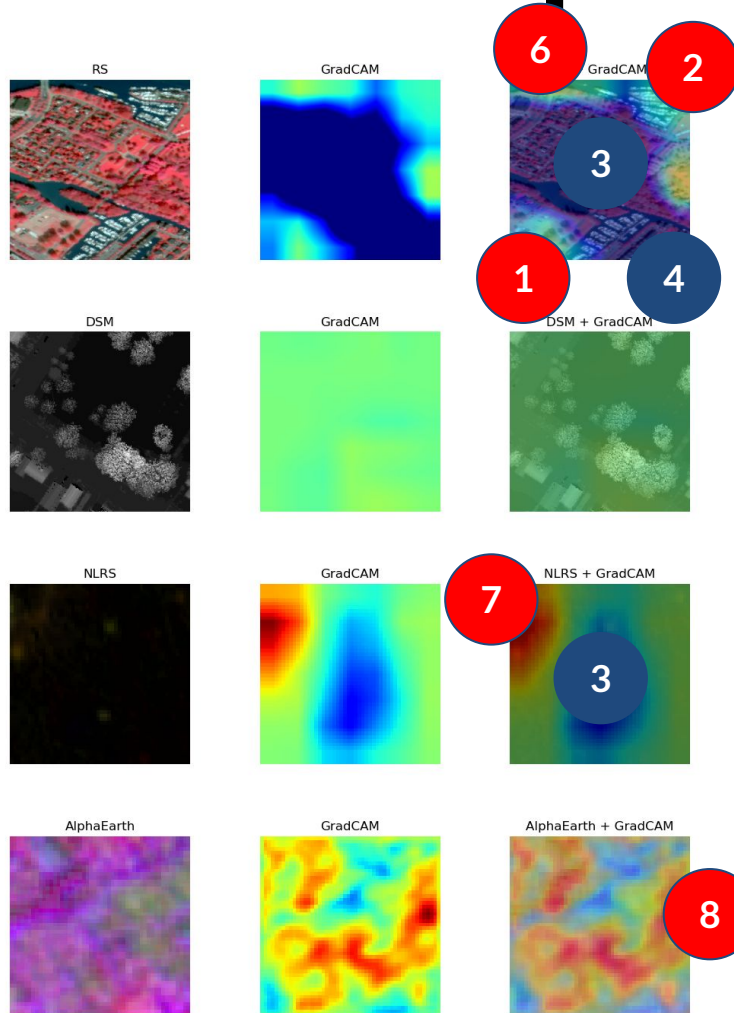
References

- Brown, Christopher F. et al. "AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data." ArXiv abs/2507.22291 (2025)
- Zhou, Wen et al. "A transformer based multi-task deep learning model for urban livability evaluation by fusing remote sensing and textual geospatial data." Remote Sensing of Environment (2026)

Baseline Grad-CAM



AlphaEarth Grad-CAM



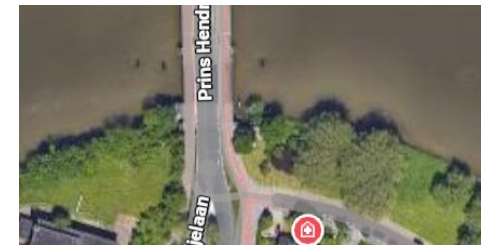
5 Not included, was unified through AlphaEarth that it was a row residential house

6 Prins Hendrikbrug: Bridge

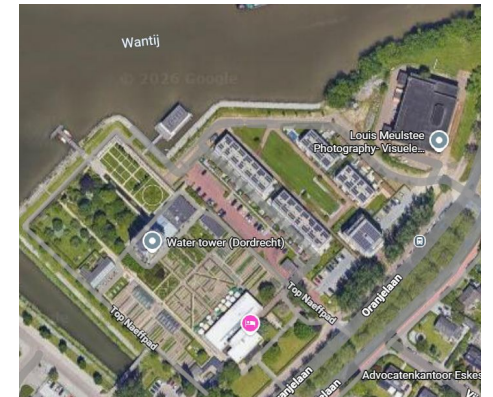
7 Dordrecht water tower: Landmark with surrounding walkable gardens

8 Green space

6



7



8

